

Clustered Index

1 What is a Clustered Index?

A **Clustered Index** is an indexing structure defined on an ordered data file where the search key field is a **Non-Key** attribute (meaning duplicate key values are allowed across records).

★ GOLDEN RULE

Clustered Index = Ordered + Non-Key (Duplicates Allowed)

Condition	Meaning in Database Context
Ordered Data	Data records are physically stored in sorted order based on the search key.
Non-Key Field	Values can repeat multiple times across different student/employee records.

2 Dataset Example

Consider an Employee Table physically sorted by **Department Number**:

Department No. (Non-Key)	Employee Name	Phone Number
1	A	111
1	B	222
1	C	333
2	D	444
2	E	555
3	F	666
3	G	777

Here, Department No. is **Sorted** ✓ but **Non-Unique (Duplicates exist)** ✗. This exact combination requires a **Clustered Index**.

3 Primary Index vs. Clustered Index

Feature	Primary Index	Clustered Index
Data Ordering	Ordered / Sorted	Ordered / Sorted
Search Key Field	Key Field (Unique)	Non-Key Field (Duplicates allowed)
Uniqueness	Strictly Unique (e.g., Roll No)	Non-Unique / Repeated (e.g., Dept No)
Index Structure	Generally Sparse	Sparse

4 How is a Clustered Index Created?

Even if search values like 1, 2, 3 repeat multiple times in the data file, we **do not create an index entry for every duplicate instance**. Instead, each unique search key value gets exactly one entry in the index table pointing to the first data block containing that value.

6 Problem of Multiple Blocks & Block Hanger

When a single non-key value (e.g., Dept 4) contains many records, those records can overflow across multiple disk data blocks. The index entry points to the first block, and subsequent blocks are chained using a **Block Hanger**.

Visualizing Block Hanger Chaining:

[Index Table] [Data Block 1] [Data Block 2]
| Key: 4 | Pointer | —> | Records: 4, 4, 4, 4 | —Block Hanger—> | Records: 4, 4, 5, 6 |

7 Search Complexity Comparison

Because index entries are sorted, we perform **Binary Search** over the index blocks (N).

Primary Index Access Cost

$\log_2 N + 1$

Unique keys guarantee that all required data resides inside exactly **one** data block.

Clustered Index Access Cost

$\log_2 N + \text{Extra Block Accesses}$

Repeated values may span across multiple blocks via Block Hanger pointers (+1, +2, +3 blocks).

8 Can We Have More Than One Clustered Index?

In standard DBMS disk organization, **a table can have AT MOST ONE Clustered or Primary Index**. This is because records can physically be ordered/sorted on disk in only *one* sequence at a time.

9 Key Exam Questions & Answers

Q1. What are the mandatory conditions for a Clustered Index?

Answer: The data file must be physically ordered/sorted, and the search key must be a Non-Key attribute (duplicates allowed).

Q2. Is a Clustered Index dense or sparse?

Answer: Sparse. One entry exists per distinct search-key value, not per individual record.

Q3. What is a Block Hanger?

Answer: A block hanger is an additional pointer mechanism used to link overflow blocks when duplicate records of the same search-key value spill into adjacent blocks.

Q4. Why might a Clustered Index require more than 1 data-block access after searching the index?

Answer: Because the search key is non-unique, multiple records matching the search value can be scattered across several consecutive disk blocks.

EXAM SUMMARY TRAP

- Primary Index = Ordered + Key (Unique) → Sparse → $\log_2 N + 1$
- Clustered Index = Ordered + Non-Key (Duplicates) → Sparse → $\log_2 N + \text{Extra Accesses}$

10-Second Revision Summary

Clustered = Ordered + Non-Key + Sparse + Block Hanger (Multi-block traversal potential)